The water Togo does not count

Drinking water access, flood risk and facility allocation · Environment Data Challenge

388 cantons mapped, and a median 26 kilometres to the nearest recorded facility. Four percent of cantons hold 35% of the exposed population, and the national inventory does not cover them.

Built by Kokou PIPI · Togo open data — opendata.gouv.tg

2 · Data, cleaning, approach

Seven resources used, one join key, and what we refused to do

Datasets loaded

6

six cleaned datasets, from seven loaded resources

Cantons in the reference

388

the corpus's only exhaustive grid

Recorded facilities

285

two disjoint inventories, never summed

TdE fields published

7 / 33

of those its dictionary describes

The join key

What links three different producers?

The canton name, normalised

No shared identifier exists across the datasets. The normalised name matches 67 of 67 TdE facilities and 191 of 218 COSO micro-projects — the remaining 27 are kept and counted, never matched at random.

Zeros are the data

What about cantons with no facility?

Left join, systematically

An inner join would erase them, and the dashboard would show a fully equipped country. It is the absence that carries the finding.

What we did not do

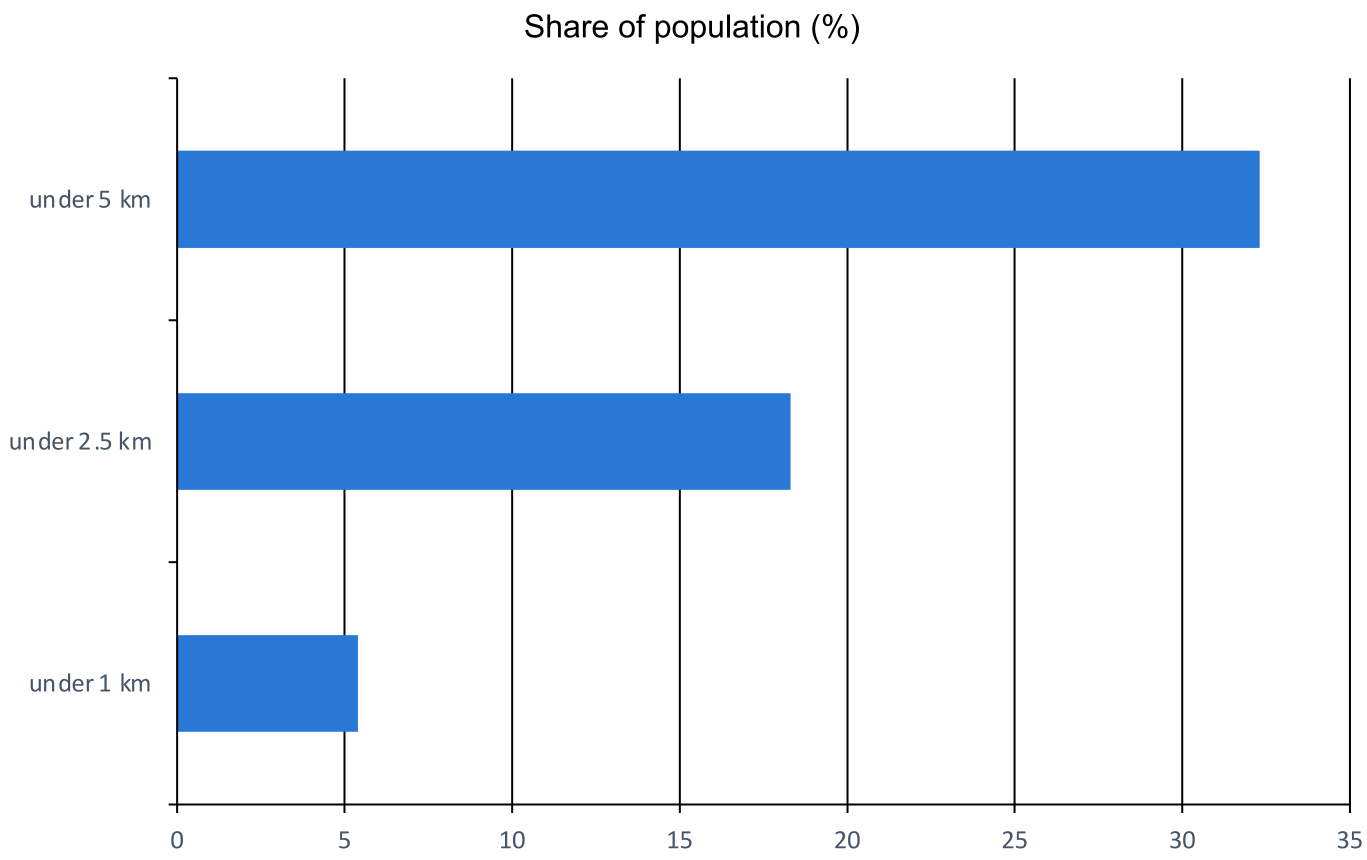
Which temptations set aside?

No national facility total

The two stocks cover disjoint territories — 97 % of TdE in Maritime, COSO in the North. Adding them would fabricate an inventory that does not exist.

3 · Where the water is, and how far away

Population reached, by distance to the nearest recorded facility



Median distance to water

26 km

from a canton to the nearest recorded facility

Cantons beyond 10 km

288

53% of the country's population

What the distance measures

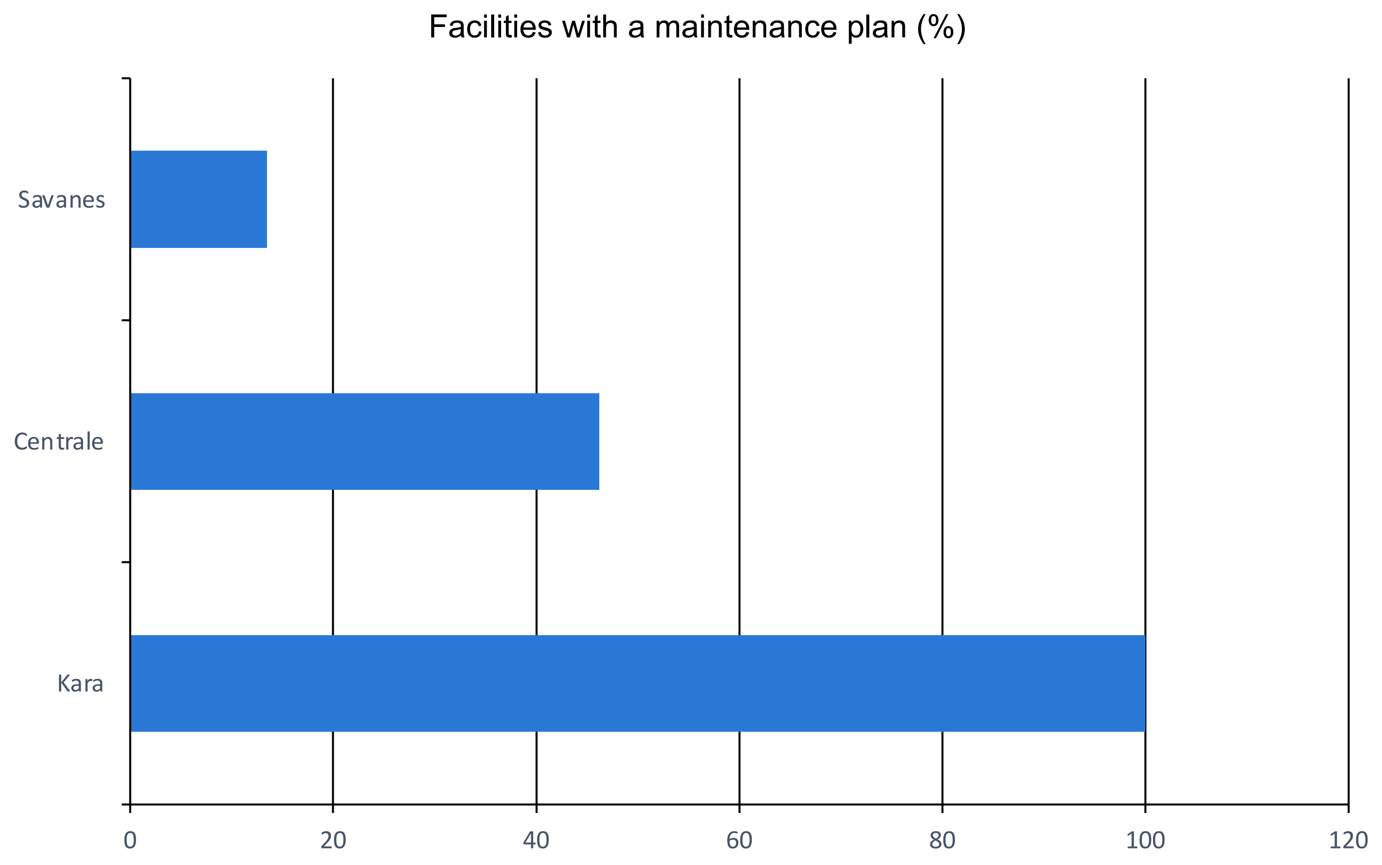
Poorly served, or poorly recorded?

R = 0.41 — facilities are clustered

The Clark-Evans index is 1 when points fall at random; below 0.5 they are clustered. So this distance measures the inventory as much as the service: a canton with no recorded facility is not a canton without water, it is a canton whose water nobody publishes.

4 · Facility condition: the missing field

No operating status in the corpus — two proxies, and what they cannot replace



Fields described, not published

26

of 33 described by the TdE dictionary

Works accepted, never handed over

74

median handover delay: 398 days

The functionality rate

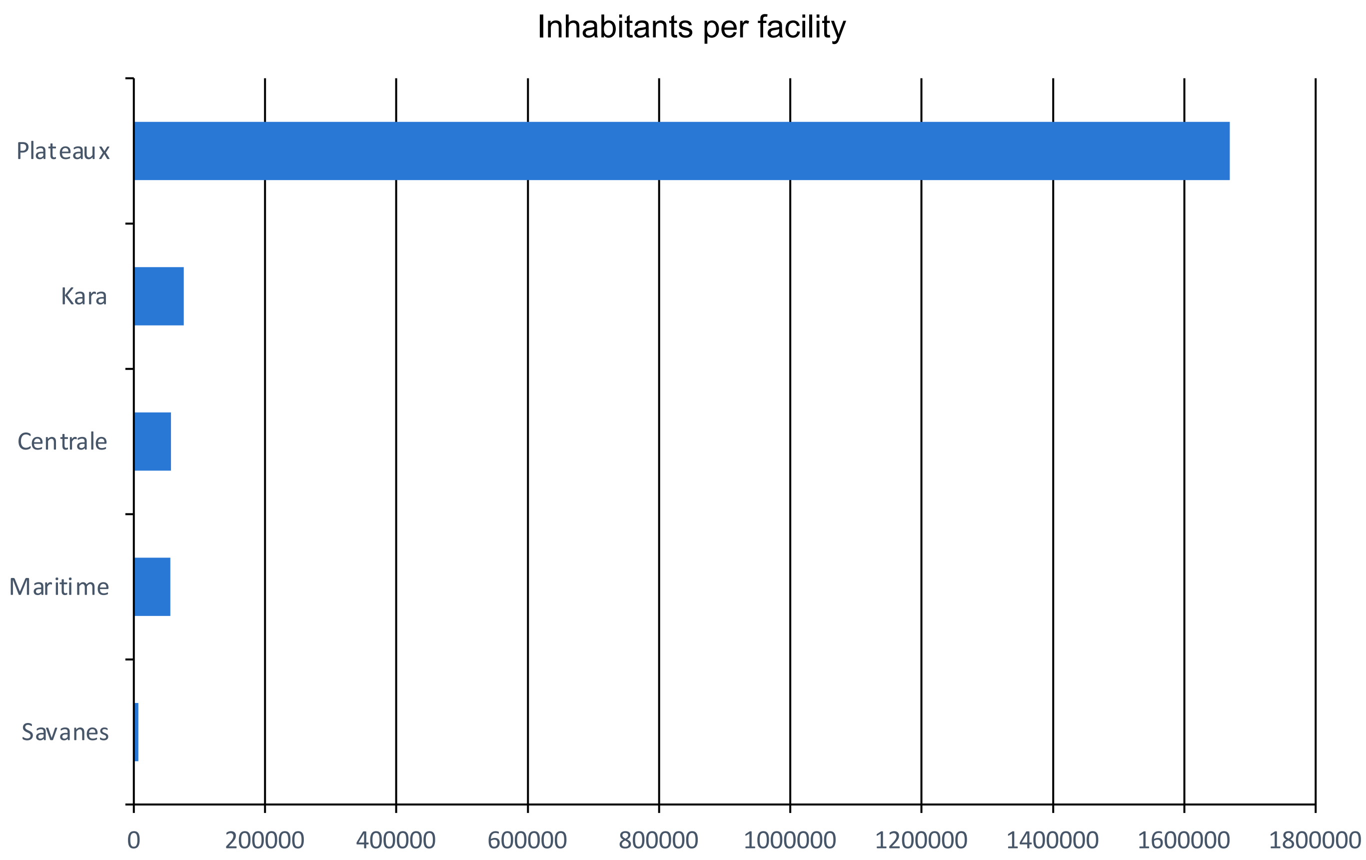
Can it be computed?

No — no field carries it

The brief asks for breakdown rates by region; the corpus cannot deliver them: no status, no failure date, no abandonment. A maintenance plan and a handover are upkeep signals, not service measures — a facility with a plan may well be down, and nothing here would say so.

5 · Demographic pressure against the inventory

Inhabitants per recorded facility, region by region



Plateaux

1.7 M

inhabitants for a single recorded facility

What the average hides

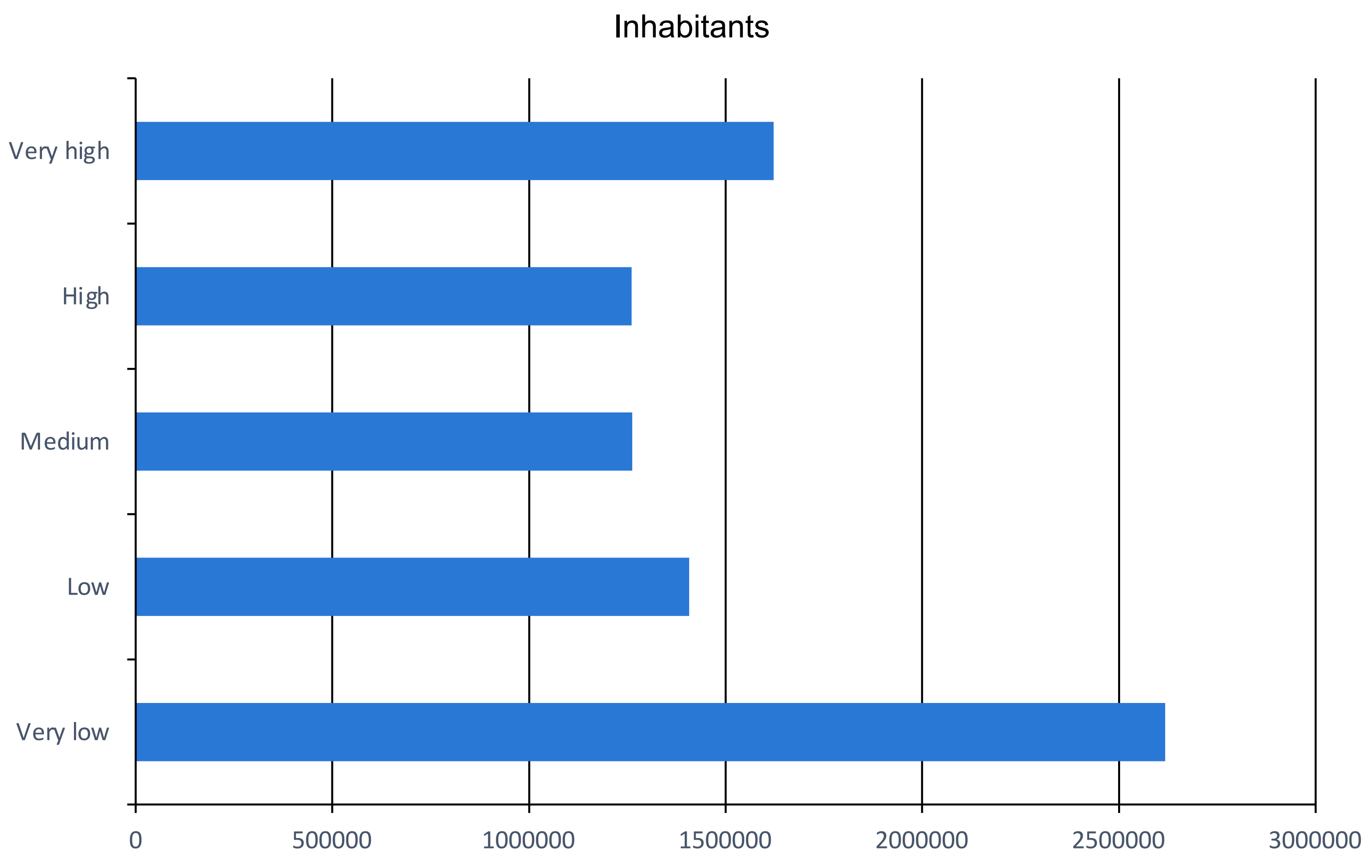
Is a national average enough?

31.5 k inhabitants per facility

It covers a 1-to-245 ratio between Savanes, where a programme concentrated its micro-projects, and Plateaux, which the inventory almost entirely ignores.

6 · Risk is concentrated; equipment is not

Population by official risk class, and what explains where facilities are built



High-risk cantons

17

4% of the grid, 2.9 M inhabitants

What explains the allocation

Need, or the programme's footprint?

7% from need, 27% adding the region

Population, risk and wealth explain little; adding the region quadruples the explanatory power. 12 exposed cantons — 1.7 M inhabitants — have no recorded facility.

The expected objection

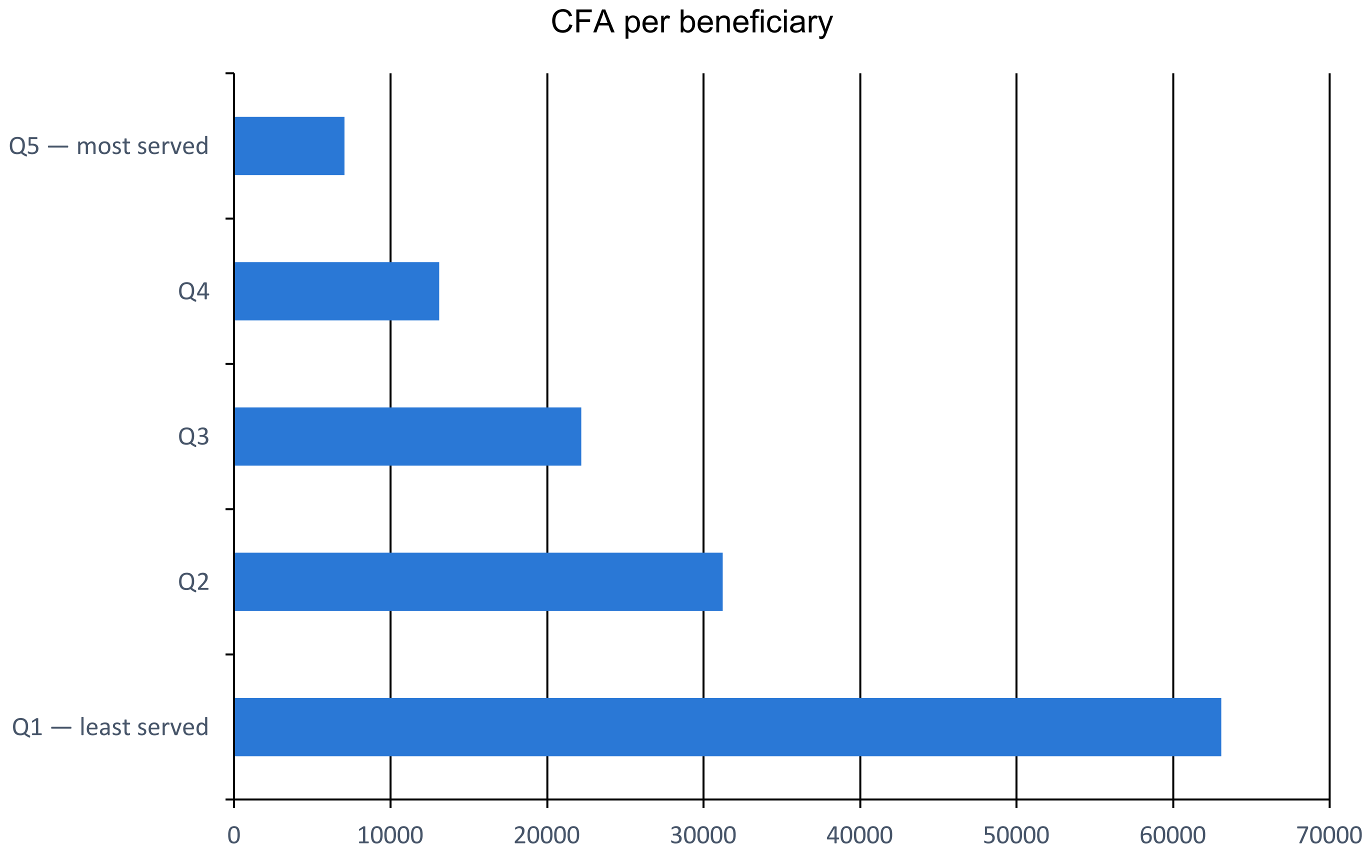
Does the index just rank populous cantons?

No — ρ = 0.94 without population

Dropping the demographic component and recomputing the index moves cantons by 3 places at the median, across the 42 where the official formula reproduces. The ranking holds without population: it does not rank people, it ranks ground.

7 · A borehole's cost does not depend on whom it serves

Cost per beneficiary, by quintile of population served



Median facility cost

16.8 M

whatever the number of beneficiaries

Elasticity to population

0.55

over 35 equipped cantons · Gini of 0.39

A decision variable

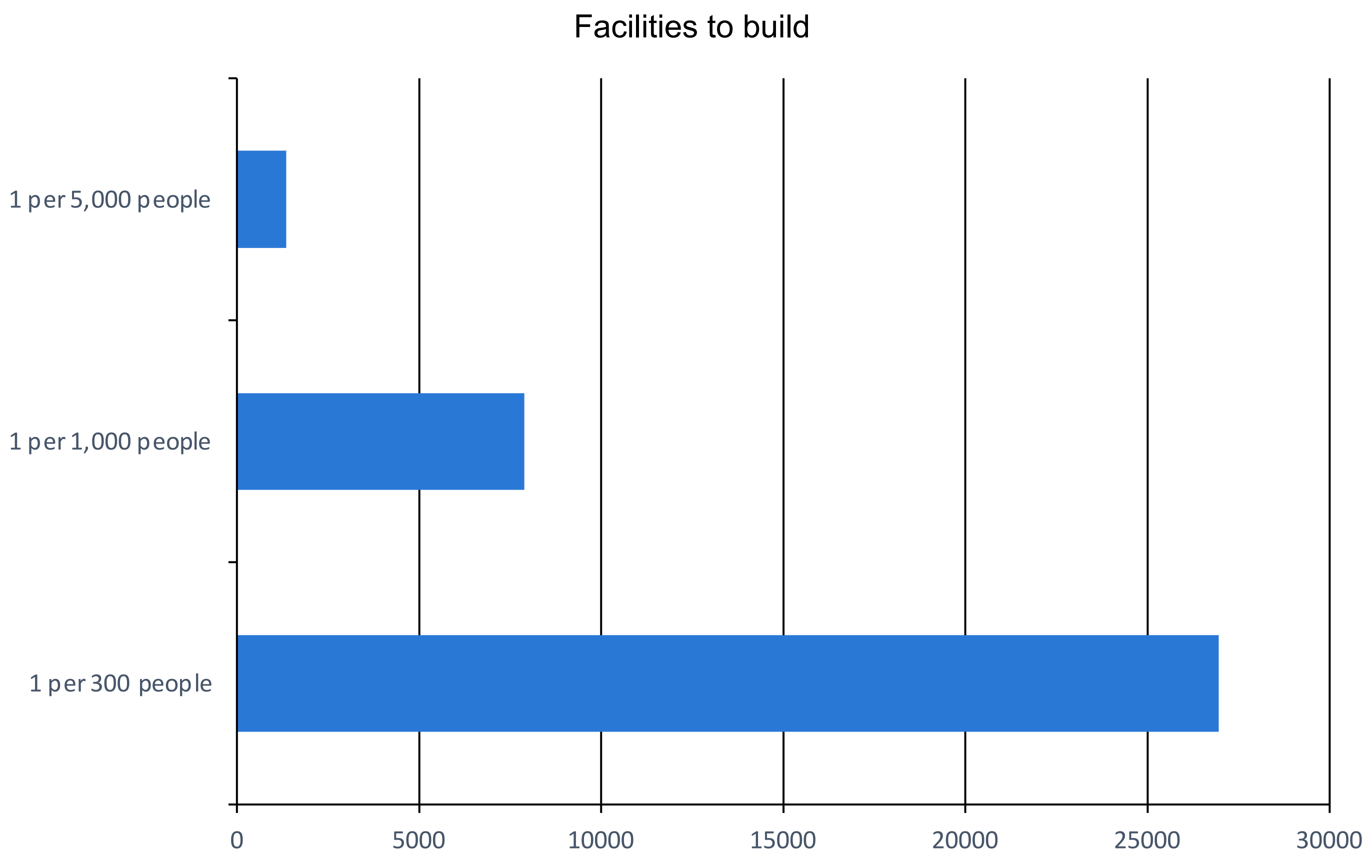
Where does the per-head cost gap come from?

9-fold, from first to last quintile

The elasticity of cost to beneficiaries is statistically nil: cost per inhabitant is therefore decided by choosing WHERE to build, not by negotiating price. Competitive tendering also came in 17% below the initial estimate, for 3.5 bn FCFA disbursed — headroom already freed.

8 · How many facilities are missing, and at what price

Three service thresholds, three programmes — the threshold is a policy choice, not a finding



Facilities to level the regions

34

onto the national median — 567.9 M FCFA

Scenario: 1 per 1,000 people

131.7 bn

7,886 facilities to build

The unit price

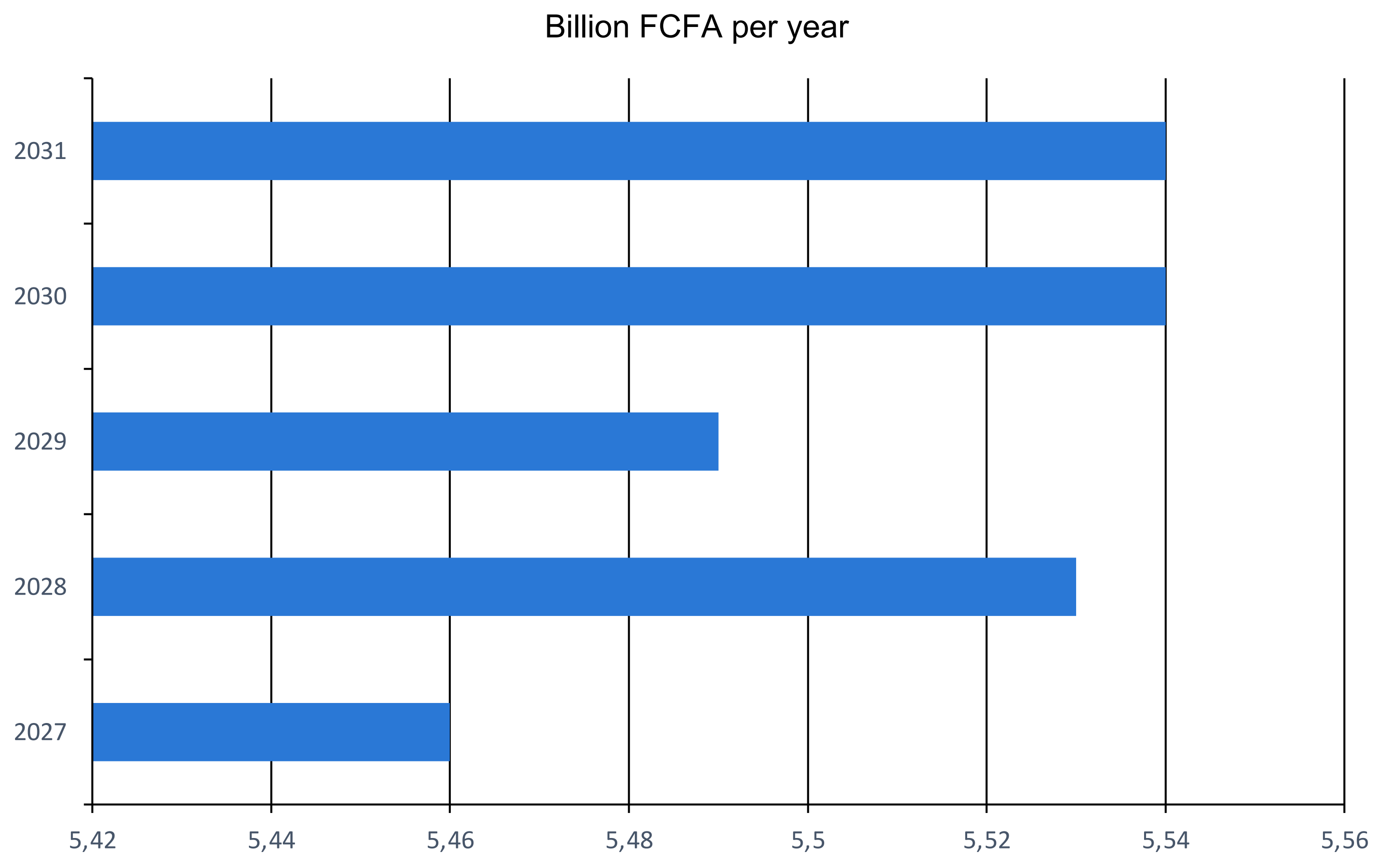
Where does it come from?

16.7 M FCFA, observed median

Median of amounts actually paid on 215 COSO facilities, not a price list. It therefore carries that programme's conditions — northern labour, borehole depth, contract year — and does not transpose as-is to another region.

9 · Two dated programmes, canton by canton

What the dashboard puts on the map: a site, a year, an amount



Facilities programme

27.6 bn

1,651 facilities over 5 years, 1 per 5,000 people

Flood programme

4.2 bn

17 cantons over 3 years · 250.0 M FCFA per works, an ASSUMED cost

Upkeep, the forgotten line

What does the stock cost once delivered?

11% of investment, at the observed rate

COSO sets aside 1.35% of a facility's cost for upkeep. At the usual three to eight percent, the cumulative charge would reach 64% of the investment by the horizon. A stock built without upkeep stops working — and stopping is exactly what the corpus cannot show.

10 · What this corpus establishes, and what it refuses

Two conclusions, three acknowledged limits

What is established

What can be stated without reservation?

Risk is concentrated, allocation does not follow it

Seventeen cantons hold a third of the country. Where facilities stand is better explained by programme footprints than by need, and the median distance to water says the same thing differently. The catch-up is costed, dated and placed canton by canton.

What remains impossible

Which stated objective is unreachable?

Functionality rates

No breakdown, abandonment or commissioning field appears in the published corpus. This is not data that was never collected: the producer's dictionary describes it. It is data collected and not released.

Three limits this report accepts

The estimates identify no causal effect — no random assignment, no discontinuity, no instrument. They are conditional associations, and their value lies in whether they survive region fixed effects.

The COSO stock covers three northern regions and the TdE stock Maritime alone: no national inference can be drawn from their execution.

The two population sources — 2010 census and 2022 satellite estimate — are not comparable term by term; their gap measures method as much as demography.